

Data Mining for Big Data

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- Assessment:
 - Evaluation on practical sessions (coeff. 1/3).
 - Written exam (2h coeff. 2/3);
- Material available on Moodle

Outline for this Part (B. Jeudy)

- ➊ Introduction and MapReduce model
- ➋ Practical Sessions (Spark / Python)
- ➌ Finding Similar Items
- ➍ Recommendation Systems

Bibliography :

- This course mostly based on part of: *Mining of Massive Datasets*, Jure Leskovec, Anand Rajaraman, Jeffrey D. Ullman.
On the website www.mmids.org, you can find:
 - the book
 - videos and slides

Important : Practical Sessions

- University computers already configured
- Can use your own computers but :
- You are responsible for having a working installation of Spark ≥ 2.3 and Python ≥ 3 **before the practical sessions**
 - Spark documentation spark.apache.org/docs/latest
 - Spark download spark.apache.org/downloads.html
 - See document in the practical part (Moodle) to test your installation
- You need to know a bit of Python Langage
 - Python documentation docs.python.org/3/index.html and specially the tutorial docs.python.org/3/tutorial/index.html
 - Read and practice the tutorial before the practical sessions if you are not familiar with python (specially the part on lists).

- Large Hadron Collider (LHC)
 - 300 GigaBytes/s of raw data
 - 15 PetaByte/year
- 30 billion pieces of content shared on Facebook / month
- 235 TeraByte (US Library of Congress, 2011)
- 5 billion mobile phones in use in 2010
- 10 ExaBytes/month: storage system sold in 2016 (from 2.5 ExaByte/month in 2013)
- 40% expected growth of data generated each year
- 5% growth in IT spending
- Giga: 10^9 , Tera: 10^{12} , Peta: 10^{15} , Exa: 10^{18}

This data must be

- Stored
- Managed
- Analyzed: this course

Different types of data

- transactional data
- texts
- graphs
- data streams
- images, videos,
- multimodal data : combine images, text, transactional data, ...

What is Data Mining

Discover patterns and models

- Valid: hold on new data with some certainty
- New: non obvious to the expert
- Useful
- Understandable: human should be able to interpret the pattern

Different kinds of data mining tasks

Descriptive methods:

- find human interpretable patterns that describe the data.
- Example: statistics, clustering, association rules

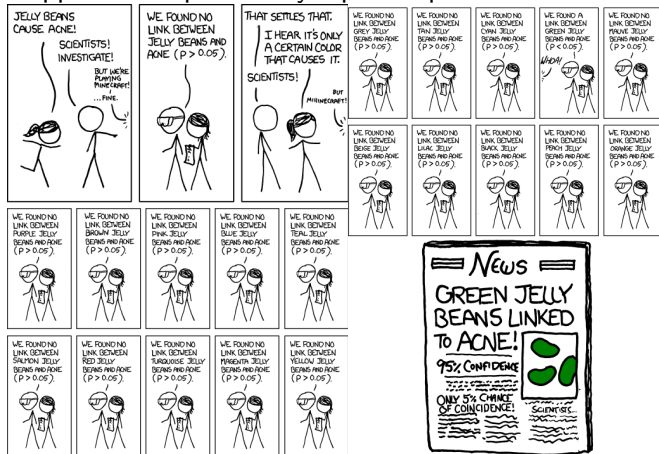
Predictive methods:

- build a model to predict some unknown/future attribute
- Example: classification, recommender system

- What happens if data is random ?
 - $\Rightarrow$ you will find patterns...
- How can we assess the quality of patterns/model ?
 - $\rightarrow$ statistical tests
 - $\rightarrow$ test set (not used to build patterns/model).

Risks of statistical tests

- In random data, a statistical test can be true by chance (false positive)
- Happens with probability equal to p-value



Bonferroni Principle

In a lot of data, there are lot of coincidences.

Finding terrorists : suspicious pairs of people

- Unrelated people who at least twice have stayed at the same hotel on the same day.
- Hypothesis:
 - 10^7 people
 - 1000 days
 - Each person stays 1 day out of 100 in a hotel (random)
 - 10000 hotels
- If everybody's behavior is random: How many pairs of such people can we expect to find?

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 - so not a good way to find terrorists...

To check if a pattern is meaningful:

- Does it happens in random data?
 - compute its expectation in random data
 - generate random data and search the pattern
- If a pattern appears in the real data and is rare in random data (happens 1% of the time)
 - $\Rightarrow$ probably meaningful
- If a pattern appears in the real data and also in random data (5000 occurrences in random data)
 - $\Rightarrow$ pattern probably not meaningful